```
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
import nltk
import random
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
nltk.download('stopwords')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
True
```

```
num_posts = 10
words = ['social', 'media', 'competitor', 'activity', 'analysis', 'keyword', 'extraction', 'experiment']

df = pd.DataFrame({
    'post_id': range(1, num_posts + 1),
    'text': [
        f"Post {i}: {' '.join(random.choices(words, k=random.randint(5, 15)))}"
        for i in range(1, num_posts + 1)
    ]
})
```

```
df.head()
```

Show hidden output

```
word_ds = []

stop_words = set(stopwords.words('english'))
```

```
def extract_key_words(text):
  words = word_tokenize(text.lower())
  keywords = [word for word in words if word.isalnum() and word not in stop_words]
  word_ds.extend(keywords)
  return ' '.join(keywords)
```

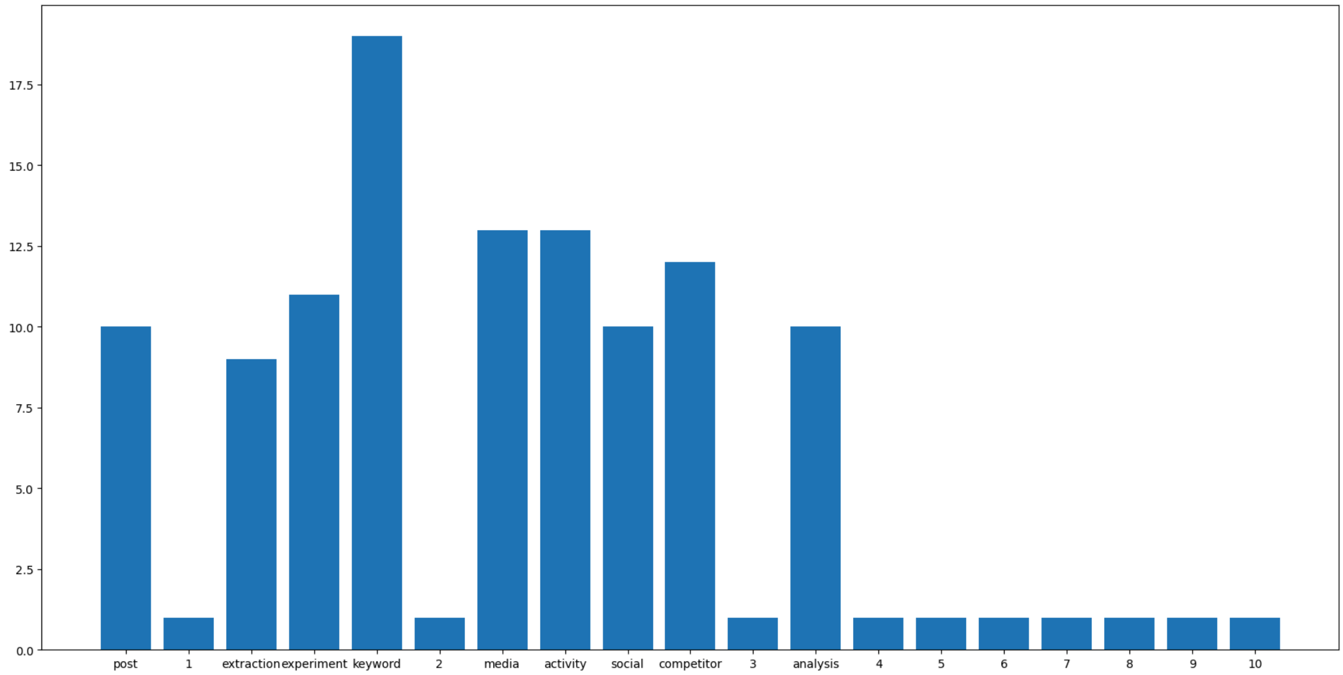
```
df['keywords'] = df['text'].apply(extract_key_words)
df.head()
```

	post_id	text	keywords
0	1	Post 1: extraction experiment extraction keywo...	post 1 extraction experiment extraction keywor...
1	2	Post 2: keyword media activity social social k...	post 2 keyword media activity social social ke...
2	3	Post 3: keyword activity media social activity...	post 3 keyword activity media social activity ...
3	4	Post 4: extraction analysis competitor activit...	post 4 extraction analysis competitor activity...
4	5	Post 5: extraction keyword keyword social extr...	post 5 extraction keyword keyword social extra...

```
counts = dict(Counter(word_ds))
print(counts)
```

```
{'post': 10, '1': 1, 'extraction': 9, 'experiment': 11, 'keyword': 19, '2': 1, 'media': 13, 'activity': 13, 'social': 10, 'competit
```

```
keys = []
values = []
for key, value in counts.items():
  keys.append(key)
  values.append(value)
plt.figure(figsize=(20,10))
plt.bar(keys, values)
plt.show()
```




```
#run again sentiment anaylysis
```